

Department of Computer Science & Information Technology



Project title:

Brain Tumor Identification based on MRI Images Using Deep Learning Techniques

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Submitted by:

Kanwal Majid BSCS/FA22/541

Zoha Rashid BSCS/FA22/520

Javeria Arshad BSCS/FA22/524

Supervised by:

Ms. Arfa Maham

Lecturer Department of CS&IT

**The Government Sadiq College Women University
Bahawalpur Pakistan**

STATEMENT OF SUBMISSION

This is certify that **Kanwal Majid** Roll no BSCS/FA22/541, **Zoha Rashid** Roll No. BSCS/FA22/520 and **Javeria Arshad** Roll No. BSCS/FA22/524 successfully completed the final project named as: **Brain Tumor Identification based on MRI Images Using Deep Learning Techniques**, at the Department of Computer Science & Information Technology, The Govt. Sadiq College Women University Bahawalpur, to fulfill the requirement of the degree of **BS in Computer Science**.

Ms. Arfa Maham

(Supervisor)

Lecturer

Department of CS&IT

Dr. Shabana Ramzan

(Chairperson)

Associate Professor

Department of CS&IT

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DECLARATION

We hereby declare that we have developed this application and accompanied report entirely based on our personal efforts. Not any of the portions of the application work presented has been submitted of any application for any other qualification or degree of this or any other university or institute of learning.

Kanwal Majid

Zoha Rashid

Javeria Arshad

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Abstract

Convolutional Neural Networks (CNNs) have revolutionized image processing, particularly in medical imaging for the identification of brain tumors, which are a major cause of cancer-related mortality. Processing complicated MRI and CT data and the lack of big annotated datasets are major obstacles. This work uses sophisticated CNN, Transfer Learning, and Vision Transformers to detect both high-grade and low-grade brain cancers. For training and assessment, an extensive dataset of CT and X-ray pictures is used. Self-attention is used by the customized Vision Transformer to recognize patterns that indicate the severity of tumors. To assess performance on criteria like accuracy, precision, recall, and F1 score, it is contrasted with models like CNN, Inception-V3, ResNet-50, Mobile-Net, and Swin Transformers. The proposed Vision Transformer model achieved exceptional accuracy of 99.61% in identifying brain tumor diseases, enhancing early diagnosis and treatment planning.

Chapter 1

Introduction

1.1 Introduction

Convolutional Neural Networks (CNNs) transformed the field of image analysis, especially in the scope of medical imaging, due to the ability to learn and detect patterns and features in images automatically. The condition of brain tumor diagnosis is common in the whole world. Brain tumors are the most common cause of cancer related death in men and women worldwide. The research problem in brain tumor identification is the processing complex MRI or CT imaging data and limitations of the lack of large annotated medical datasets. In this study, the main purpose to resolve this issue by focusing on the recognition of two common brain tumor disease classes, high grade and low grade by state-of-the-art CNN, Transfer Learning and Vision Transformers (ViT) techniques. We train and test on a large dataset of CT and X-ray images. The Vision Transformer with self-attention architecture is carefully designed to capture complex patterns indicative of high grade and low grade. A specialized Vision Transformer with self-attention architecture is developed and compared with CNN, Inception-V3, ResNet-50, Mobile-Net and Swin Transformers models to evaluate its effectiveness in identifying intricacy of patterns related to high grade and low grade brain tumor disease. The objective of this thesis is to develop and recognize DL algorithms for proper detection and classification of brain tumors from MRI images, thus improving early diagnosis planning for better patient outcomes.

This study describes the comparison to typical contrast-enhanced PET/CT lung tumor setting up, the use of CT in a paired single-step procedure outlined here is possible and incorporates physio-pathological data regarding the biological processes of lungs cancer related to tumor circulation and digestion, without the need for the use an extra contrast agents as well as an acceptable greater radiation exposure for the individual. We look at recently suggested the most advanced deep learning architecture and methods as CAD systems for the identification of lung illness. They fall separated into two groups: (1) False positive decrease systems, which categorize a collection of given prospective tumors into cancerous or benign cancers; and (2) Nodule identification systems, which identify candidate nodules from the original CT scan. The primary attributes of the various methodologies are showcased, and their efficacy is scrutinized. Additionally,

introduced are the research-useful CT lung datasets. An assessment and discussion of the various strategies are provided.

1.2 Project Title

Brain Cancer Identification based on MRI Images Using Deep Learning Techniques

1.3 Project Overview Statement

This project aims to use the power of (CNNs) and ViTs in image processing, along with the effectiveness of transfer learning to address the challenges of limited and diverse datasets used in scanning medicine. The research aims to provide a reliable, automated and easily available tool for brain tumor detection by combining these cutting-edge technologies. The goal is to help medical practitioners, enhance patient results and perhaps increase the accessibility of advanced detecting tools in various healthcare settings. This synergistic combination of sophisticated AI techniques is poised to revolutionize brain tumor diagnostics, and represents a significant stride in bridging the gap between technological innovation and clinical implementation in the healthcare sector.

1.4 Project Goals and Objectives

1. Propose advanced deep learning architectures or techniques for brain cancer detection in MRI images focusing on improving the accuracy, robustness, and interpretability of these models.
2. To implement the processing efficiency of the model, ensuring it can analyze medical images, such as MRI quickly and effectively.
3. To evaluate state-of-the-art DL architectures for feature extraction and classification of brain tumor MRI images, enhancing diagnostic accuracy and efficiency.

Literature Review

Brain Tumor Diagnosis

Diagnosing a brain tumor typically involves a multi-step process:

Medical History and Physical Examination: A thorough medical history and a neurological examination are the first steps in the process. The eye, hearing, balance, coordination, strength, and reflexes are all examined by the doctor.

Imaging Tests: The most common imaging tests for diagnosing brain tumors are (MRI) and (CT) scans. MRIs distinguishing between healthy and diseased tissue. CT scans are quicker and can reveal bleeding and bone abnormalities that might be associated with brain tumors.

Biopsy: This can sometimes be done during surgery to remove the tumor, or as a separate procedure.

Additional Tests: Other tests might include electroencephalogram (EEG) to check for abnormal brain waves (often used if seizures are a symptom), spinal tap (lumbar puncture) to examine cerebrospinal fluid, and blood tests to provide information about the individual's overall health and detect problems that might indicate cancer.

Recent advancements in medical technology and advanced genomic testing, have improved the accuracy of brain tumor diagnosis, allowing for more personalized and effective treatment plans.

Deep Learning in Brain Tumor Identification

Deep Learning using large datasets to teach algorithms—usually neural networks—how to execute certain tasks. Deep learning techniques, in particular (CNNs), have demonstrated extraordinary performance in brain tumor identification.

Convolutional Neural Networks (CNNs): CNNs are the most popular deep-learning models for image recognition applications. They are extremely good at finding patterns in images, making them ideal for detecting anomalies like brain tumors. CNNs can learn to distinguish between tumorous and healthy brain tissue by analyzing thousands of training images.

Transfer Learning: Given the complexity of brain scans and the lack of labeled medical pictures, transfer learning has become a prominent method. It entails using a pre-trained model (a neural network trained on a broad, general dataset such as ImageNet) and fine-tuning it with a specific set of brain scans. This technique requires substantially less brain imaging data and computational resources.

Model Training and Validation: The prepared dataset is used to train the deep learning model, which will learn to recognize and classify brain tumors. The model's performance is then verified against a second collection of images to check its correctness and capacity to generalize to new, previously unknown data.

Advanced Techniques and Models: In addition to basic CNNs, advanced models and approaches like as U-Net for segmentation and ViTs for better feature extraction are increasingly being investigated for brain tumor detection.

This study [1] proposed a deep residual region-based CNN named Res-BRNet which is a combination of spatial and boundary feature extraction for brain tumor classification. The model was trained and evaluated on the Kaggle, BR35H, and Figshare MRI datasets, achieving 98.22% accuracy with an F1-score of 0.9841. The results demonstrate that CNN based architectures are effective in extracting local features such as edges, textures, and tumor boundaries.

This work [2] presented ShallowMRI, a lightweight CNN with a novel attention mechanism for efficient classification of brain tumors. The proposed model was tested on three benchmark MRI datasets and achieved an impressive accuracy of 99.70% in classification with low computational complexity. Such architecture is appropriate for implementation on resource-constrained devices.

The evaluated various pre-trained CNN and transformer models on a 4-class brain MRI dataset with a total of 7023 images by performing 5-fold cross validation [3]. The Swin-Tiny Transformer outperforms various traditional CNN backbones, and among all the evaluated models, it has the highest accuracy of 99.24%. The study emphasizes the effect of feature learning using transformer for classification of brain tumors.

This study [4] have introduced an improved attention mechanism and dual-branch down-sampling Swin Transformer V2 for brain tumor classification. With a 7,023-image grayscale MRI dataset, the model obtained an accuracy of 98.97%, which is better than the popular CNN models like ResNet50 and DenseNet121. The findings of the study showed the superiority of the advanced architectures of transformer in medical image analysis.

This study [5] proposed a hybrid network that was built by using Swin Transformer and ResNet50V2 in conjunction with transfer learning and extensive data augmentation. The proposed model attained 99.9% accuracy in the binary classification and 96.8 % accuracy in the classification of four brain tumors. The hybrid method proved to be successful in fusing CNN with transformer-based global feature learning.

A transfer learning study [6] was carried out to assess the performance of various pre-trained deep CNN architectures on a 4-class MRI dataset comprising 7023 images. The Xception model was superior in terms of classification accuracy, resulting in a maximum weighted classification accuracy of 98.73%, which was more than the other popular CNN models. The results indicate the transfer learning approach is an effective approach for medical image classification.

To enhance brain tumour classification, Murala and Rao suggested the use of a Vision Transformer (ViT) model with MRI image segmentation [7]. The model was based on self-attention mechanisms to model long-range dependencies and global contextual information from MRI images. Classification results on Kaggle four class dataset showed competitive results.

Fine-Tuned Vision Transformer (FTVT) models were introduced by [8] for the classification of four classes brain tumor with glioma, meningioma, pituitary tumor, and normal MRI images. All tumor types benefited from fine-tuning, resulting in substantial improvements in feature representation and classification. The study proved transformer-based transfer learning to be effective in medical imaging.

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The Swim Transformer, inspired by the concept of Self-Attention Mechanism, introduces spatial information into the transformer architecture, enhancing its capability to capture intricate patterns within images. Conversely, the Vision Transformer, originally developed for natural image classification tasks, has showcased remarkable adaptability in medical imaging by efficiently leveraging self-attention mechanisms to learn representations directly from pixel values, eliminating the need for handcrafted features. When integrated with CNN architectures, these transformer-based models offer a potent framework for lung disease detection, facilitating comprehensive analysis of chest radiographs or CT scans.

suggested a Hierarchical Multi-Scale Attention Vision Transformer using a multi-resolution patch embedding of image patches of size 8×8 , 16×16 and 32×32 . It improved the accuracy of the classification task and significantly reduced the training time by about 35% compared to the conventional Vision Transformers. Learning efficiency and robustness were improved by multi-scale feature extraction.

In a CNN–Vision Transformer meta-ensemble built with ten pre-trained CNN and ViT models, a meta-learning classifier, and CLAHE preprocessing was introduced [12]. The ensemble achieved very good accuracy and explanation of brain tumor classification in a four-class MRI dataset. The study showed that ensemble learning can enhance the robustness and the generalization.

This research [13] suggested using a geometric-average ensemble of EfficientNetV2 and ViT-B16 for brain tumor classification. The ensemble further reduced the number of classification errors and enhanced the overall prediction performance as compared to

EfficientNetV2-M performed by itself with 98.01% accuracy. It is found from the results that the use of CNN and transformer model improves the reliability of the classification process.

In the upgraded hybrid CNN-Vision Transformer model with hybrid feature selection, Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO) are used [14]. The ViT + ResNet architecture is able to obtain the highest classification accuracy of 98.84%. The optimization procedures enhanced the feature selection and model performance in brain tumor MRI classification.

In this study, a combined model experiments were carried out on two public datasets, BraTS 2018 and Figshare, revealing that our model attained an average accuracy of 96.75% on the BraTS 2018 dataset and 99.10% on the Figshare dataset. Additionally, our suggested model surpasses various leading-edge methods on both BraTS 2018 and Figshare datasets, improving accuracy by 3.06% and 1.06%, respectively. In this research, we introduce SwinBTS, a novel approach to 3D medical image segmentation. We utilized an enhanced Transformer module, specifically developed to improve the extraction of detailed features.

The goal of segmenting images is to identify the exterior contour and voxel data associated with the region of interest. Segmenting organs or lesions is the primary use of division in vision in medicine. This makes it possible for the quantitative analysis of pertinent clinical factors and the provision of further information for follow-up diagnosis and therapy. Target separation, for instance, is essential for tumor radiation guiding and operative image navigating.

This study through the application of the LSTM Profound Learning computation, the information on the presence of individuals is inferred. Using multidimensional time series data as indicators for the relapse defining issue, a test is applied in an office. The results gained show that the framework enables the collection, connection, and preservation of environmental data. The LSTM was computed using the collected data and evaluated against other AI systems. The computations are Supporting SVM Classifier, Guileless Bayes Structure, and Inter Perceptron Feed-Forward Organization. The results that followed parametric changes show that LSTM performs better in the

indicated application. Recent technological breakthroughs like AI and deep learning have been significant in offering fixes for issues in the clinical field.

This study Investigates the efficiency of a collective of conventional Vision Transformer (ViT) models in identifying (MRI). We utilized pretrained and subsequently fine-tuned ViT models for the task of classification. The L/32 model emerged as the most proficient single model, recording a test accuracy of 98.2% with a resolution of 384×384 . By amalgamating all four ViT models, the ensemble exhibited an enhanced total test accuracy of 98.7% at this resolution, outstripping the efficacy of each individual model across both resolutions and their collective outcomes at a reduced resolution of 224×224 . In our research, we introduced an enhanced vision transformer model, dubbed FT-ViT, which leverages sophisticated image processing and deep learning strategies to precisely detect brain tumors in input images. The Contrast Enhanced Magnetic Resonance Imaging (CE-MRI) with 5712 brain tumor images, it demonstrated a high capability in accurately identifying the tumors. The FT-ViT model achieved an impressive accuracy rate of 98.13%. The process begins with effective data preprocessing, succeeded by the utilization of a (CNN) to derive key features from the images. The suggested model demonstrates a notable ability to predict brain tumors, achieving a classification accuracy of 99.1%, a precision of 98.8%, a recall of 98.9%, and an F1-measure of 99.0%.

1.5 High-Level System Components

1.5.1 Data Acquisition module

Collects MRI brain images from datasets, including tumor and non-tumor, and makes sure they are appropriately identified for training.

1.5.2 Data preprocessing module

Enhances model performance by scaling, normalizing, and applying augmentation to clean and prepare normal images.

1.5.3 Feature extraction

Automatically extracts significant characteristics from images using CNN or pretrained models.

1.5.4 Model training module

Train the deep learning model on MRI data to identify tumor patterns using frameworks like TensorFlow.

1.5.5 Model evaluation module

Evaluates the model's performance using metrics like as F1-score, accuracy, precision, and recall.

1.6 List of Optional Functional Units

- Image enhancement module
- Tumor segmentation module
- Multi-class classification module
- API integration module
- Visualization dashboard module

1.7 Application Architecture

Client side: Allows users to upload MRI images and request tumor analysis.

Server side: Processes uploaded MRI images and returns prediction results.

1.8 Gantt Chart: Brain Tumor Classification

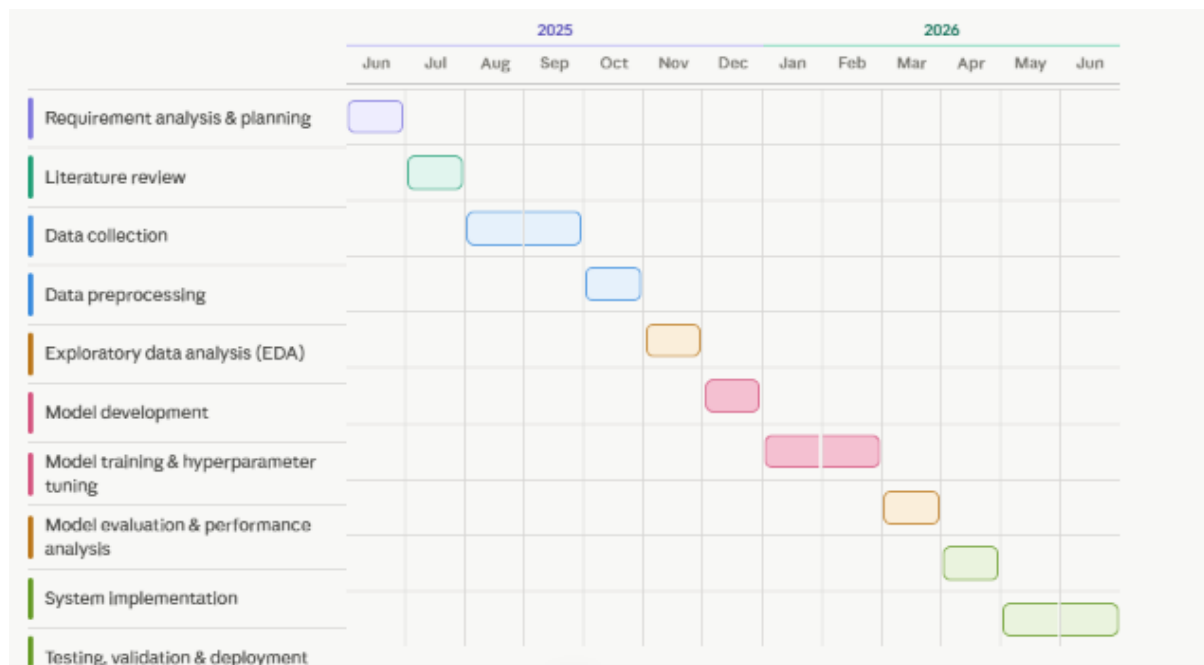


Table 1 Gantt chart of brain tumor classification

Phase	Tasks	Duration
Project Planning	Topic selection, requirement analysis, literature review	Month 1
Data Collection	Collect MRI datasets, data labeling, dataset organization	Month 2–3
Data Preprocessing	Image resizing, normalization, augmentation, data cleaning	Month 4
Exploratory Analysis	Dataset analysis and feature visualization	Month 5
Model Design	Select CNN/Transfer Learning models and define architecture	Month 6
Model Development	Implement CNN and transfer learning models	Month 7
Model Training	Train models and tune hyperparameters	Month 8–9
Model Evaluation	Accuracy, Precision, Recall, F1-Score, ROC analysis	Month 10
Comparative Analysis	Compare suggested model with existing models	Month 11
Testing & Validation	System testing, bug fixing, validation	Month 11
Documentation & Thesis Writing	Results discussion, thesis writing, revisions	Month 12

1.9 Hardware and software specification

Table 2 Detail of hardware and software specifications

Component	Specification
Processor	Intel Core i5 or higher
RAM	16 GB
Software Stack	Python, TensorFlow, Anaconda
Operating System	Windows / Linux / Mac
Browser	Chrome, Firefox

Chapter 2

Feasibility Study

2.1 Introduction

The brain tumor classification solution is checked to see if it will work. To make sure the brain tumor classification project is done well we look at it from sides, including the technical, operational, economic and ethical sides of the brain tumor classification project.

2.2 Project Feasibility Report

2.2.1 Technical Feasibility

We have deep learning frameworks and image processing technologies so the brain tumor classification project is technically possible. We can use frameworks like TensorFlow to build and train neural networks for medical image processing. We have the MRI datasets we need to train the models. We can get them from public sources. We also have image processing programs like scikit-image and OpenCV to help with things like normalization and augmentation. If we have computers and resources like GPUs we can train and use models to detect and classify brain tumors.

2.2.2 Operational Feasibility

The brain tumor classification system we suggest is easy to use so it is operationally possible. We can upload MRI images. See the results and it is not hard to use, even for people who are not experts in technology. The system helps with diagnosis and makes it faster and easier so we do not have to do much work by hand.

2.2.3 Economic Feasibility

The brain tumor classification system is not expensive because we use open-source tools and frameworks like TensorFlow and OpenCV. We do not have to pay a lot of money for software. We can use regular computers or cloud-based resources, which saves us money on hardware. This makes the brain tumor classification solution affordable for medical projects and research, about brain tumor classification.

2.1.1 Schedule Feasibility

A structured timeline was followed with the following major milestones:

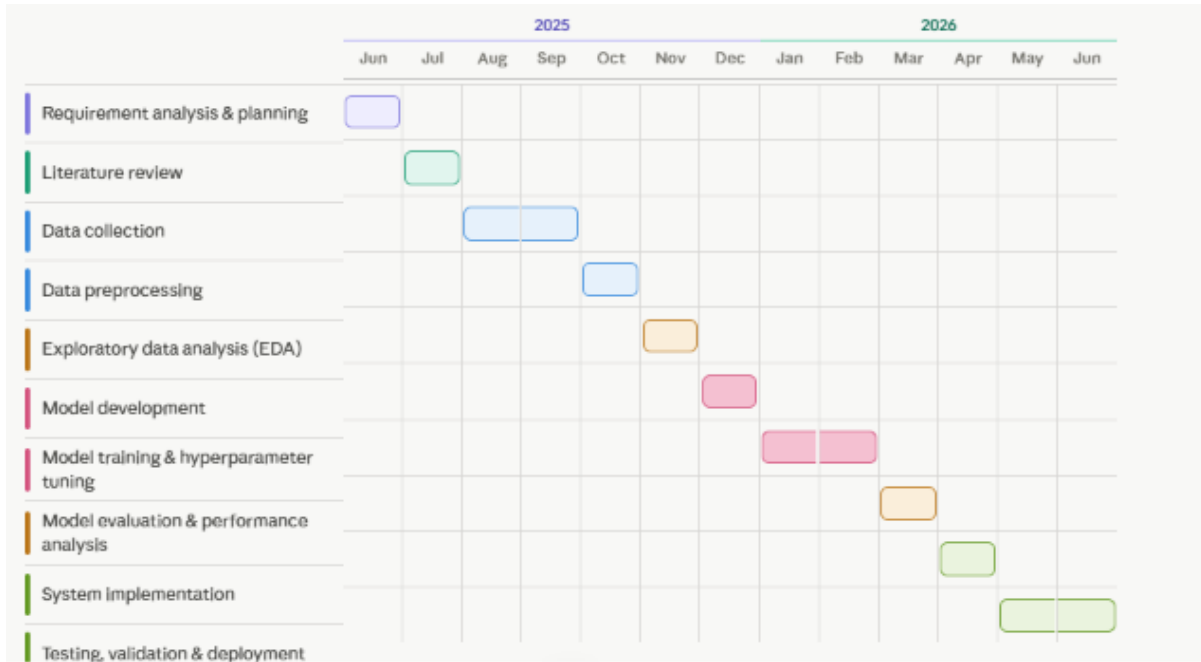


Table 3 Structured timeline

Tasks	Estimated Duration
Requirement Analysis & Planning	Month 1
Literature Review	Month 2
Data Collection	Month 3–4
Data Preprocessing	Month 5
Exploratory Data Analysis (EDA)	Month 6
Model Development	Month 7
Model Training & Hyperparameter Tuning	Month 8–9
Model Evaluation & Performance Analysis	Month 10
System Implementation	Month 11
Testing, Validation & Deployment	Month 12

2.1.2 Specification Feasibility

- Requires standard hardware (16 GB RAM, GPU for faster training)
- Supports common operating systems (Windows, Linux, Mac)
- Uses widely available tools (Python, TensorFlow, OpenCV)
- Compatible with modern web browsers

-
- Scalable for future enhancements

2.1.3 Information Feasibility

The MRI image datasets used by the system are accessible from both public and medical research sources. Accurate model training requires proper labeling and high-quality data. The system makes sure that medical picture data is handled, processed, and stored for analysis in an efficient manner.

2.1.4 Legal and Ethical Feasibility

The system must handle patient MRI data in accordance with ethical and medical data privacy regulations. To preserve patient identification, appropriate permission and data anonymization should be guaranteed. Furthermore, rather than taking the role of expert medical judgment, the system should be utilized as a helpful diagnostic tool.

2.2 Project scope

The thesis primarily focuses on integrating traditional deep learning models with the emerging Vision Transformer (ViT) technology, enhanced through transfer learning strategies.

- Develop a system to detect cancer from MRI images using DL techniques.
- This integration objectives to capitalize on the strengths of each method, offering a more accurate and efficient approach for identifying brain tumors from medical imaging.
- Provide a simple interface for uploading images and viewing results.
- The dissertation emphasizes the efficacy of transfer learning in improving model performance, especially when dealing with limited medical data.
- Visualize model performance using graphs and evaluation metrics.

2.3 Task Dependency Table

Table 4 Dependency table

Task	Dependent On
Requirement Analysis	None
Data Collection	Requirement Analysis

Data Preprocessing	Data Collection
Model Design	Data Preprocessing
Model Training	Model Design
Model Evaluation	Model Training
System Implementation	Model Evaluation
Testing & Deployment	System Implementation

2.4 Vision Document

The project's goal is to develop an good and effective system that helps with brain tumor early identification and categorization. The technology seeks to decrease the time needed for manual MRI image interpretation and increase diagnosis accuracy by utilizing robust DL models.

The objective of this project is to create a dependable and easy-to-use tool that can aid researchers in making decisions. The system will produce precise predictions and visual insights using contemporary frameworks like TensorFlow and PyTorch. By incorporating AI-based technologies into medical imaging procedures, the ultimate objective is to improve healthcare outcomes.

2.5 Project Features

2.5.1 Automated Tumor Detection

The method uses trained deep learning models to automatically identify brain tumors from MRI scans, saving manual work and increasing speed.

2.5.2 Tumor Classification

It allows for more thorough investigation and diagnosis by grouping cancers into many categories (such as glioma, meningioma, and pituitary).

2.5.3 Image Preprocessing

The system preprocesses MRI images through resizing, and cleaning to improve model accuracy and performance.

2.5.4 Model Training and Evaluation

This helps DL models be trained and assesses them using performance metrics such as recall, accuracy, and precision.

2.5.5 Visualization of Results

Shows graphs like accuracy and loss curves to help users in understanding the evaluation of the model.

Chapter 3

Requirement Analysis

3.1 Introduction

The project needs to figure out what is required. This means looking at what the system needs to work. We have to think about what the system can do. For example, the system needs to be able to upload MRI images. It also needs to be able to preprocess these images. Then it has to identify tumors and classify them. The system also needs to show the results in a way that's easy to understand. These are the things the system needs to do. There are also other things to think about. The system needs to work. It needs to be accurate. It needs to be easy to use. It needs to be dependable. It needs to work on different platforms. We also have to think about things like TensorFlow. This is a deep learning framework. We need to think about what hardware and software. We need computers that are powerful enough to train and use the models. The project needs things like MRI images and tumor identification and classification. The project also needs things, like system performance and accuracy and usability.

3.1.1 System Specification

Hardware requirements: To train deep learning models more quickly, the system needs a computer with an Intel Core i5 CPU or above, 8GB of RAM, and optional GPU support.

Software requirements: The project is developed in Python and uses image preparation tools such as OpenCV, NumPy, and scikit-image. Models are developed using DL method such as TensorFlow.

Operating System Requirements: The system is interoperable with all major operating systems, including Windows, Linux, and macOS, providing flexibility throughout development and deployment.

Dataset Requirements: For model training, validation, and testing, the system requires labeled MRI brain imaging datasets that include both tumor and non-tumor images.

3.1.2 Existing System

Radiologists manually evaluate MRI images to determine the existence and type of brain tumor. This technique is primarily dependent on medical knowledge and can be time-consuming, particularly when evaluating a large number of images. Existing methods

may also use basic image processing techniques with less automation and poorer accuracy than deep learning systems.

Limitations:

- Manual diagnosis is time-consuming and requires expert radiologists.
- Human error may occur due to fatigue or subjective interpretation.
- Traditional methods have limited automation and scalability.
- Difficulty in handling large MRI datasets efficiently.

3.1.3 Scope of the System

The scope of suggested system is:

- To detect brain tumors from MRI images using deep learning techniques.
- To preprocess MRI images for improving model performance.
- To classify images into tumor and non-tumor categories.
- To provide automated prediction results with minimal human effort.
- To display model performance using graphs and evaluation metrics.
- To offer a simple interface for MRI image upload and result visualization.

3.1.4 Identify External Entities

Below are the identified external entities and their interaction with the system:

Table 5 External entities of the system

Entity	Description	Action
User	Person who interacts with the system to analyze MRI images.	Uploads MRI images and views prediction results.
Admin	Manages the system and monitors model performance.	Updates datasets, manages system settings, and monitors outputs.
MRI Dataset Source	Source of labeled MRI brain image datasets used for training and testing.	Provides MRI images for preprocessing and model training.

Deep Learning Model	Trained model used for tumor detection and classification.	Processes MRI images and generates predictions.
----------------------------	--	---

3.1.5 Context Level Data Flow Diagram (DFD)

A high-level summary of the Brain Tumor Identification System's interactions with outside parties may be seen in the Context Level Data Flow Diagram (DFD). It shows the complete system as a single process that uses the taught deep learning model to process the user's MRI images. After analyzing the supplied data, the system produces prediction results that are returned to the user. For data management and model changes, the system communicates with external entities like the dataset source, and storage module. Understanding the general information flow between the system and external components is made easier by this figure.

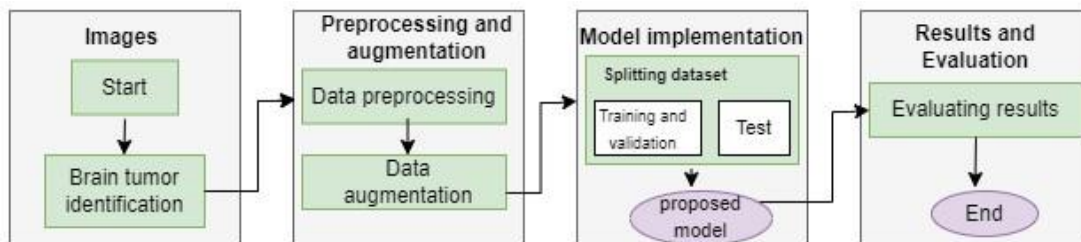


Figure 3-1 Flowchart diagram of brain tumor classification

3.1.6 Capture "Shall" Statements

These are requirement statements expressed as obligations:

- The system shall allow users to upload MRI brain images for analysis.
- The system shall preprocess MRI images by resizing, normalizing, and enhancing them.
- The system shall use a DL model to detect the presence of brain tumors.
- The system shall generate prediction results with probability scores.
- The system shall display results in a user-friendly interface.
- The system shall store MRI data, trained models, and prediction outputs.

3.1.7 Allocate Requirements

Table 6 Allocated requirements of the system

Module	Allocated Functionality
Data Input Module	Upload MRI images for processing
Preprocessing Module	Resize, normalize, and enhance images
Feature Extraction Module	Extract features using CNN or pretrained models
Model Training Module	Train deep learning model on MRI dataset
Prediction Module	Classify MRI images as tumor or non-tumor
Visualization Module	Display graphs and prediction results
Storage Module	Store datasets, models, and results

3.1.8 Prioritize Requirements

Table 7 Prioritize Requirements

Priority	Requirement
High	MRI image classification and tumor detection accuracy
High	Image preprocessing and model prediction functionality
High	Model training and evaluation using deep learning
Medium	User-friendly web interface for image upload and results
Medium	Visualization of results and performance metrics
Low	Optional features like report generation and dashboard

Low	Model retraining and advanced analytics features
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3.2 Summary of Requirements (Initial Requirements)

3.2.1 Functional Requirements

- The system **shall** allow users to upload MRI brain images for analysis.
- The system **shall** preprocess MRI images by resizing, normalization, and enhancement.
- The system **shall** detect the presence of brain tumors using a deep learning model.
- The system **shall** classify MRI images into tumor and non-tumor or tumor types.
- The system **shall** display prediction results with accuracy or probability scores.

3.2.2 Non-functional Requirement

- The system **shall** ensure high accuracy in tumor detection and classification.
- The system **shall** provide fast response time for image detecting and prediction.
- The system **shall** have a user-friendly and simple interface for ease of use.
- The system **shall** be scalable to handle large MRI datasets efficiently.
- The system **shall** ensure data security and safe handling of medical images.

3.2.3 Requirements Traceability Matrix

Table 8 Requirements traceability matrix

Requirement ID	Description	Module
FR1	MRI image upload	Input Module
FR2	Image preprocessing steps	Preprocessing Module
FR3	Tumor detection process	Prediction Module
FR4	Tumor classification system	Classification Module
FR5	Result display output	UI Module

NFR1	High model accuracy	Model Evaluation
NFR2	Fast processing speed	System Performance
NFR3	User-friendly interface	Frontend UI
NFR4	Scalable system design	System Architecture
NFR5	Data security handling	Security Module

Chapter 4

System Analysis and Design

4.1 Introduction

The design phase is concerned with the architectural, structural, and component-level design of the brain tumor classification. This contains use case diagrams, sequence diagrams, and system-level architecture, which show how each module interacts as data flows across the system.

4.2 High Level System Use Case Diagram

This use case shows what happens when the user and the system work together. The user and the system have some interactions and the system and the user are important to understand. The user and the system are the parts of this so we will look at the user and the system.

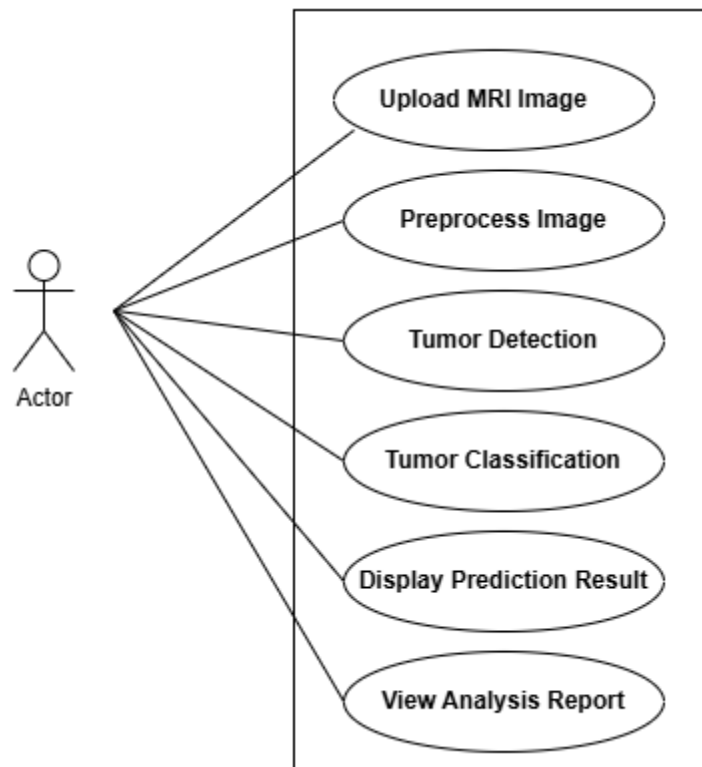


Figure 4-1 High level system use case diagram

Actor:

User (Upload image)

Use case:

- Upload MRI Image
- Preprocess Image

-
- Tumor Detection
 - Tumor Classification
 - Display Prediction Result
 - View Analysis Report

4.3 System Sequence Diagram

4.3.1 Admin Sequence Diagram

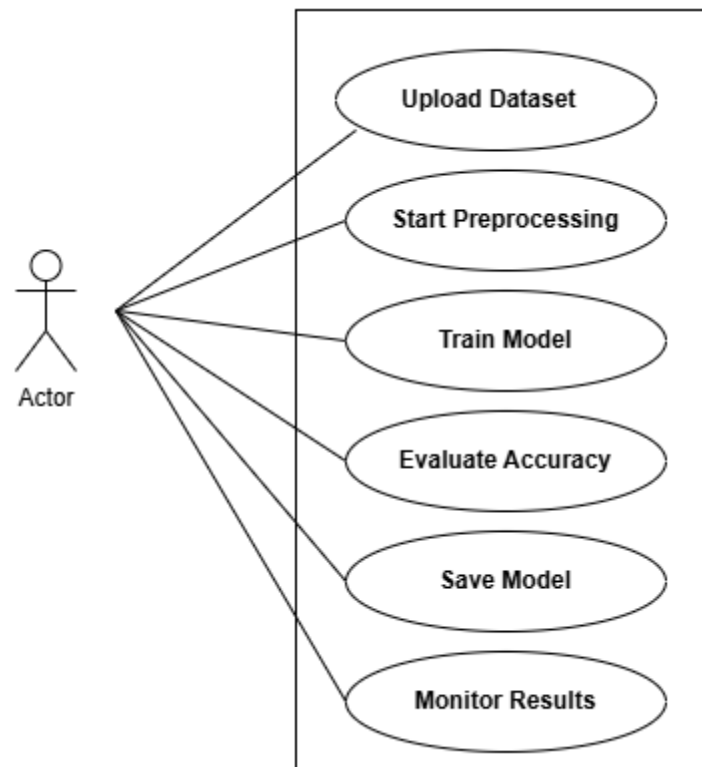


Figure 4-2 Admin Sequence diagram

Actor:

Admin (Upload Dataset)

Use case:

- Upload Dataset
- Start Preprocessing
- Train Model
- Evaluate Accuracy

- Save Model
- Monitor Results

4.3.2 User Sequence Diagram

User → Open System

→ Upload MRI Image

→ Request Analysis

System → Preprocess Image

→ Run Deep Learning Model

→ Detect / Classify Tumor

System → Generate Prediction Result

→ Display Output

User → View Result

4.4 Collaboration Diagram

User → System Interface

System Interface → Preprocessing Module

Preprocessing Module → Deep Learning Model

Deep Learning Model → Prediction Module

Prediction Module → Result Display

Result Display → User

4.5 Operation Contracts

Table 9 Operation contracts for MRI-Brain tumor

Operation	Precondition	Post condition
Upload MRI Image	User is logged in and MRI image is available	MRI image is uploaded successfully to the system
Image Preprocessing	MRI image must be uploaded	Image is resized, normalized, and prepared for analysis
Tumor Detection	Preprocessed image is available	System detects presence or absence of tumor
Tumor Classification	Tumor is detected or model is ready for classification	Tumor type is classified successfully
Generate Prediction Result	Analysis process is completed	Prediction result with score is generated

View Results	Prediction result is available	User can view output and analysis details
Store Data	Prediction or model output is generated	Results and data are saved in storage

Chapter 5

Implementation of MRI Brain Tumor

5.1 Introduction

This part explains how we use a kind of artificial intelligence to identify brain tumors from MRI images.

We outline the dataset we used

- * The architecture of our model
- * How we prepared the images
- * Techniques we used to make images
- *. How we measured its performance.

It serves as a guide for implementing and validating our approach.

This research looks at how CNN and transfer learning can help identify brain tumors, which's very important in medical imaging and diagnosis.

- With learning CNN has become a powerful tool for analyzing complex medical images.
- This offers ways to accurately detect and classify brain tumors.
- Using transfer learning makes our model better by using -trained networks on large datasets.
- This helps solve the problem of not having medical imaging data.
- Our goal is to improve how well CNN architectures and transfer learning work together.
- We want to make brain tumor identification more accurate, efficient and reliable.
- This will help improve diagnosis methods in healthcare.

5.2 Dataset

The brain tumor disease identification dataset has a lot of images. These images are used to identify two types of brain cancers: grade and high grade. The dataset has two classes: High-Grade and Low-Grade tumors. The High-Grade class has 3453 images of brain tumors that're more aggressive. The Low-Grade class has 3221 images of brain tumors that're less aggressive and grow slower. There are images of one type of tumor than the other. This shows that brain tumors can be very different. The dataset is good for testing learning and

Vision Transformer techniques to identify brain tumors automatically. The brain tumor disease identification dataset is useful, for creating and testing these techniques because it has a lot of images of brain tumors.

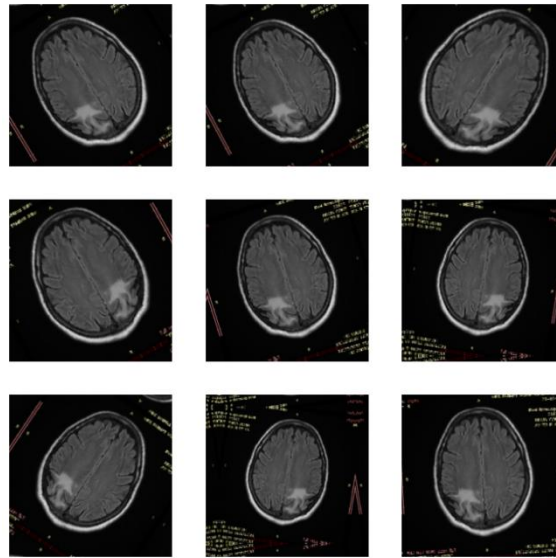


Figure 5-1 Dataset Sample

5.3 Data Preprocessing

The first part of this work is preprocessing. The main goal of preprocessing is to make images look better which helps with putting things into categories and separating them. Preprocessing the data is an important step when you are working with machine learning. It includes things like making images normalization adding more data removing the background and making the colors look more even. It also includes dealing with missing information encoding labels and splitting the data into parts to make it easier to analyze. This step is very important for any machine learning project because the way the data is set up and how good it is will affect how well the model works.

Preprocessing helps get rid of noise and complicated things in the data set so the ML algorithms can focus on the important parts. This means that the system can use its resources efficiently which makes the model train and make predictions faster. Preprocessing also helps the system learn more about the connections and trends in the data, which makes it

better, at identifying brain tumors as you can see in the figure. Brain tumor identification is what preprocessing is helping with. The preprocessing of the data is making brain tumor identification more accurate and reliable.

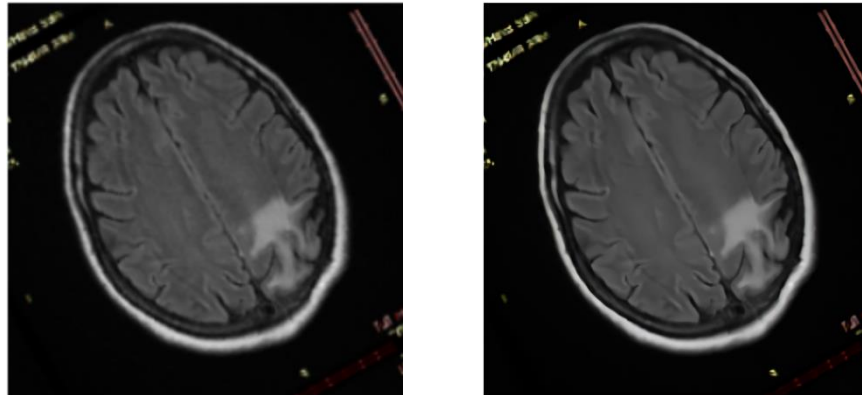


Figure 5-2 Preprocessing

(a) Original image

(b) Preprocessed image

5.3.1 Image Resizing

The Data collection is separate. Data is divided into verification and instruction sets. Originally the images were really big they were 240x320 pixels. The images have been resized to 224x224 pixels to make them better, for image classification tasks. This is what people usually do for image classification tasks because it makes it easier to compute and train Convolutional Neural Network models. Resizing the images to a size is very useful.

5.3.2 Normalization

The normalization of brain cancer recognition using CNN involves making sure the model works accurately. This process helps in identifying and classifying lung diseases. To do this we adjust images to a fixed size of 224x224 pixels. This way all images have the size no matter what their original size. We then change the pixel values to make them easier to work with. Each pixel value is divided by 255. This makes the pixel values range from 0 to 1. The goal is to make the image pixel intensities consistent. The output is. Pixel intensities range from 0 to 255. This helps the model to work with images. The normalization process is

crucial, for lung disease detection. It ensures that the model works with images of sizes. The Convolutional Neural Networks can then classify lung diseases accurately. The process involves steps to optimize the models outcomes. The model can then be used to detect lung diseases.

$$I_{new(x,y)}^{normalized} = \frac{I}{\sum_{j=-1}^1 \sum_{i=-1}^1 1} \sum_{j=-1}^1 \sum_{i=-1}^1 1 * I_{old}(x+i, y+j) \quad (1)$$

5.3.3 Data Augmentation

Using data really helps CNN find brain cancer. This is a part of making them work better. To do this we make lots of pictures, from the original pictures of leaves. We use things like turning the pictures around flipping them over and making them bigger or smaller to make the group of pictures we use to train the Convolutional Neural Networks interesting. This helps the Convolutional Neural Networks learn from the pictures and then be able to find brain cancer in pictures it has not seen before. the pictures bigger or smaller is when we change the size of the picture so some parts are bigger and some parts are smaller. The Convolutional Neural Networks can then look at the pictures. Find brain cancer. This technique can help create variations of the original images and improve the model's object identification capability at different scales. Horizontal Rotating entails rotating the image vertically to produce a reflected copy of the source. Some tasks include in data augmentation are as following:

5.3.4 Data Splitting

Dividing the dataset into training and testing sets is a critical step in the development and testing of a CNN model used for identifying lung disease. It assists to partition the image data into individual sets for training and testing. The data slicing technique is employed to divide the images into training and testing sets in this study. The test percentage is 0.2, this indicates that 20% of the images are used for testing. These images are not included during the model training process. The remaining 80 percent of images are utilized for training of CNN model. In training, the model is taught the patterns in the lung images and the ability to predict improves. Once trained, the test images test the performance of the model. This process will indicate the accuracy of the CNN in recognizing lung

disease in an image that it has never seen before. A data split offers assessment and assists in decreasing false results. It also aids in the comprehension of the accuracy, reliability and performance of the model.

5.4 CNN Architecture

In this study, we present a CNN-based algorithm for detecting brain tumors. The simulation parameters for the suggested CNN structure are detailed in Table. For image classification, a CNN utilizes TensorFlow and Keras. The design of the system processes input images measuring 224x224 pixels in three color channels (RGB). The input layer is defined with a shape of (224, 224, 3), indicating the expected dimensions of the input images. Data enhancement involves transforming the input photos, such as rotating, flipping, and zooming, in order to add more diversity to the training data.

The next includes a rescaling layer, which scales the input photos' pixels correspond to a range between 0 and 1. This is a typical stage of preparation to normalize the input data. The architecture of the model includes several sequences of convolutional layers, where each is succeeded by batch normalization, max-pooling, and activation functions. The convolutional layers (Conv2D) are in charge of extracting characteristics through the application of filters to the input images. The ReLU Non-linearity is introduced to initiate the function model, and batch normalization helps stabilize and accelerate the training process.

After each convolutional block, a max-pooling layer is implemented to minimize the spatial dimensions and retain essential features. This down sampling helps the model focus on the data while discarding less relevant details.

The last part of the architecture includes a flattened layer to reshape the 2D feature maps into a 1D vector, then an expulsion layer having a 0.5 dropout rate. During training, dropout is used to randomly deactivate a portion of neurons in order to prevent overfitting. The outcome of a binary classification task is indicated by the last layer, which is a thick layer with two output data. Using a softmax stimulation function, probabilities scores for every class are generated, and the model is compiled using an appropriate loss function, optimizer, and evaluation metrics.

Table 10 Parameters detail of CNN architecture

Parameters	Values
Convolutional layer	6 (3*3)
Pooling layer	3 (3*3)
Optimizer	Root Mean Squared Propagation (rmsprop)
Epochs	20
Batch size	32
Learning rate	0.001

5.5 ResNet-50

Integrating ResNet-50 into a sequential model for brain tumor disease identification, with a focus on proves to be a potent strategy. This study constructs a ResNet50 CNN, which has been pre-trained, is used as an extractor of features in a deep-learning model for brain tumor diagnosis. The model architecture follows a transfer learning approach, utilizing the understanding obtained from training on the ImageNet dataset for a different task.

The initial part initializes the ResNet50 a model on the ImageNet dataset that has already been weighted. The completely upper connecting layers are not represented by the `include_top=False` option of the network, and the `input_shape=(224, 224, 3)` specifies the expected shape of the input images, which are assumed to be 224x224 pixels with three color channels (RGB).

Following the initialization of ResNet50, a Keras Sequential model is created. ResNet50 is added as the second layer in the sequential model, serving as a feature extractor. ResNet50 is known for its deep architecture with residual connections, allowing it to capture hierarchical features in images effectively. In the context of brain tumor identification, these features might include patterns and abnormalities associated with various lung conditions.

A layer that drops out with a rate of dropouts of 0.25 is introduced after the ResNet50 layer. Dropout is a regularization method that enhances the model's capacity to generalize to new data by preventing overfitting and randomly deactivating a portion of neurons during training. The model architecture is further extended with dense layers. The feature maps extracted by ResNet50 are flattened, and two dense layers with rectified linear unit (ReLU) activation functions are added. These dense layers act as classifiers, transforming the extracted features into a representation suitable for binary classification.

The ResNet50-based model has a total of 8 layers, as shown in table 3.2. This count includes the layers for data augmentation, the ResNet50 convolutional base, dropout layers; flatten layers, and dense layers for feature transformation and classification.

Table 11 Parameters detail of ResNet_50

Parameters	Values
Layers	9
Optimizer	adam
Epochs	30
Batch size	32
Dropout	0.25
Learning rate	0.001

5.6 MobileNet

The inclusion of MobileNet in the neural network model for brain tumor identification serves specific purposes in the context of classification tasks. MobileNet is chosen for its efficiency, lightweight architecture, and the ability to strike a balance between accuracy and computational cost. The model incorporates MobileNet as its convolutional base, and various layers are added to the model for feature extraction, transformation, and classification.

The model's initial layer adds is the data augmentation layer, denoted by `data_augmentation`. This layer is essential to increase the generalization ability of the simulation by using arbitrary rotations to the input images during training. Data augmentation helps improving the model's robustness to variations in input data.

The second layer, `'conv_base'`, represents the MobileNet convolutional base. MobileNet is especially made for portable, effective deep learning algorithms applications, making it suitable for limited in resources environments such as integrated and portable gadgets. The convolutional base is typically pre-trained using substantial information sets, such ImageNet, and is capable of extracting hierarchical features from images.

Following the dropout layer, the model is flattened using `layers.Flatten()`. This operation transforms the output from the convolutional base to convert it into a one-dimensional vector so that it can be fed into the dense layers that come after. Two dense (fully connected) layers with Rectified Linear Unit (ReLU) activation functions are then added to the model. These layers, with 256 and 128 units, respectively, perform feature transformation and abstraction on the flattened output. ReLU activation.

Another dropout layer is introduced after the first dense layer with a dropout rate of 0.25. This additional dropout layer further helps prevent overfitting during the training process. The final dense layer has two output nodes with a softmax activation function (`layers.Dense (2, activation='softmax')`). This indicates that the model is configured for binary classification, predicting the probability of an input belonging to one of two classes. In the context of brain tumor identification, these classes could represent healthy and diseased lungs. Table summarizes the parameter detail of MobileNet, which describes layers, epochs, batch size, optimizer, and learning rate.

Table 12 Parameters detail of MobileNet

Parameters	Values
Layers	8
Optimizer	Root Mean Squared Propagation (rmsprop)
Epochs	25

Activation function	ReLU
Batch size	32
Learning rate	0.001

5.6 Inception V3

Deep convolutional neural network architecture called InceptionV3 has shown to be successful in image categorization applications. It is distinguished by the use of inception modules, which use several convolutional filters operating in parallel at different dimensions to collect characteristics at different scales. The model can be optimized for particular applications like brain tumor identification and is frequently already trained on huge datasets like ImageNet.

The InceptionV3 model is added as the convolutional base. It consists of multiple inception modules, each capturing features at different levels of abstraction. The pre-trained weights of InceptionV3 enable the model to recognize intricate patterns and structures within the lung images. To combat overfitting, a dropout layer with a rate of 0.25 is incorporated following the convolutional base. It functions by intermittently disabling a portion of the neurons throughout the training process, thereby enhancing the model's ability to generalize. Additionally, a flatten layer is employed to convert the output of the InceptionV3 convolutional base into a one-dimensional vector, making it suitable for feeding into the subsequent dense layers.

An additional dropout layer, with a rate of 0.25, follows the initial dense layer to aid in regularization throughout the training. The concluding dense layer includes two output nodes with a softmax activation function, configuring the model for binary classification. This setup allows the model to estimate the likelihood of an input image being categorized into one of two groups, indicative of either healthy or affected lungs. Table summarizes the parameter detail of MobileNet, which describes layers, epochs, batch size, optimizer, and learning rate. The model is designed to learn and discern lung patterns, contributing to accurate classification of lung diseases.

Table 13 Parameters detail of InspectionV3

Parameters	Values
Layers	7
Optimizer	Root Mean Squared Propagation (rmsprop)
Epochs	30
Batch size	32
Learning rate	0.001

5.7 Swin Transformer

The adoption of Swin Transformer in the context of brain tumor identification is motivated by its unique features and capabilities that align with the intricate characteristics of medical images associated with these conditions. Swin Transformer, known for its effectiveness in capturing long-range dependencies and hierarchical features, provides an ideal solution for discerning these intricate patterns within the lung images.

The characteristics in brain images can affect analysis; data augmentation ensures the model identifies strong features that remain consistent despite these changes. The Swin Transformer, a transformer-based structure, is adept at recognizing long-range dependencies in images. Its capabilities in self-attention, multi-head attention, and hierarchical analysis render it highly effective for image classification tasks. By incorporating Swin Transformers, the model can identify complex patterns and details within lung images, aiding in the identification of minor anomalies indicative of lung conditions.

Patch-Based Processing: The Patch Extract and Patch Embedding layers separate the input images into chunks before integrating them in a lower-dimensional environment. This patch-based approach permits the model to detect both local and global dependencies in lung images more efficiently. This method is very beneficial when dealing with medical photos, as certain regions of interest may hold vital diagnostic information.

Swin Transformer: The Swin Transformer is a transformer-based design that excels at detecting long-term dependencies in images. Its self-attention processes, multiple-head

attention, and hierarchical processing make it ideal for image categorization tasks. Swin Transformers enable the model to recognize intricate patterns and characteristics within lung images, which aids in the diagnosis of minor anomalies linked with lung diseases.

Global Average Pooling: The Global Average Pooling1D layer expands information from the final set of patches into a single-dimensional vector. This method lowers the dimensions of space equal to each feature's single value, readying the data for the final classification layer. Global average pooling is a common method for decreasing data while keeping significant properties.

Dense Output Layer: The model concludes with a dense layer that contains two output nodes and a Softmax activation function, indicating that it is set up for binary classification. This layer forecasts the probability distribution of the input falling into one of two categories, most likely representing healthy or ill lungs.

The table indicates that architecture starts with an input layer that accepts photos with a specific input form. The Patch extract and patch embedding layers separate the input image into patches and embed them in a lower-dimensional space. Two Swin Transformer blocks follow, each representing a layer of the Swin Transformer model, which is particularly adept at capturing long-range dependencies and hierarchies in images. The Patch Merging layer joins neighboring patches, allowing for interactions between adjacent regions.

Table 14 Parameters detail of swin transformer

Parameters	Values
Layers	10
Optimizer	Adam
Epochs	10
Batch size	32
Learning rate	0.001

5.8 Proposed Vision_transformer with Self-attention

The proposed methodology focusing on the identification of brain tumor disease, introduces an innovative approach leveraging transformer-based models with self-attention mechanisms. In this methodology, the input brain tumor images undergo a series, including random cropping and flipping to augment the dataset and enhance the model's ability to generalize across diverse clinical scenarios. The unique aspect of the methodology lies. The brain tumor images are initially processed through a PatchExtract and Patch Embedding layer, breaking them down into patches and embedding them into a lower-dimensional space. Two instances of the Swin Transformer, each representing a distinct block of the model, follow suit. The Swin Transformer's self-attention mechanisms are crucial in enabling the model to analyze relationships between distant regions within the brain tumor images, providing the capability to discern subtle patterns associated with carcinoma. The proposed methodology incorporates patch-based processing, allowing the model to focus on local features while maintaining the global context, a crucial aspect in the nuanced detection of lung abnormalities.

The suggested model for diagnosing brain tumor diseases utilizes a Vision Transformer (ViT) architecture, demonstrating considerable effectiveness in tasks related to image classification. The model begins by taking input images of a specified shape and applies data augmentation to enhance its generalization capabilities. Following augmentation, the images undergo a patching process, breaking them down into patches using a Patches layer. These patches are then encoded through a Patch Encoder, which projects them into a lower-dimensional space. The process of patch extraction for lung disease detection involves resizing an image to 224x224 pixels and extracting 256 patches, each containing 588 elements is shown in figure. This technique is commonly used in image-based tasks like lung disease identification and detection, as it captures local patterns and features, enhancing the detection model's sensitivity to subtle brain tumor disease patterns.

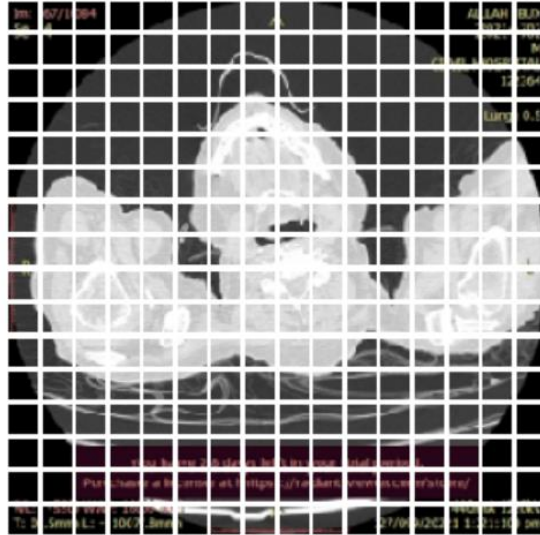


Figure 5-3 Patch extraction

The core of the model consists of multiple Transformer blocks, each comprising a series of operations. Within each block, Layer Normalization is applied first, followed by a multi-head self-attention mechanism, allowing the model to capture intricate dependencies between patches, as shown in figure. A preserve important information from the original encoded patches. The block continues with layer normalization, followed by a Multi-Layer Perceptron (MLP) for capturing non-linear features. Another skip connection integrates the MLP's output with the previous attention output.

This process is repeated for the specified number of Transformer layers, improving the model's performance to capture both local and global features in the lung images. After the Transformer blocks, the encoded patches undergo additional layer normalization, flattening, and dropout before entering an MLP head for further feature extraction. The final classification layer consists of a dense layer with a softmax activation function, producing probabilities for binary classification.

The transformer layers and MLP head units define the depth and complexity of the model architecture, as shown in table. The input shape represents the dimensions of the input images. These parameters collectively shape the behavior and performance of the proposed ViT-based lung disease detection model. Adjusting these values based on the dataset characteristics and task requirements is essential for achieving optimal results.

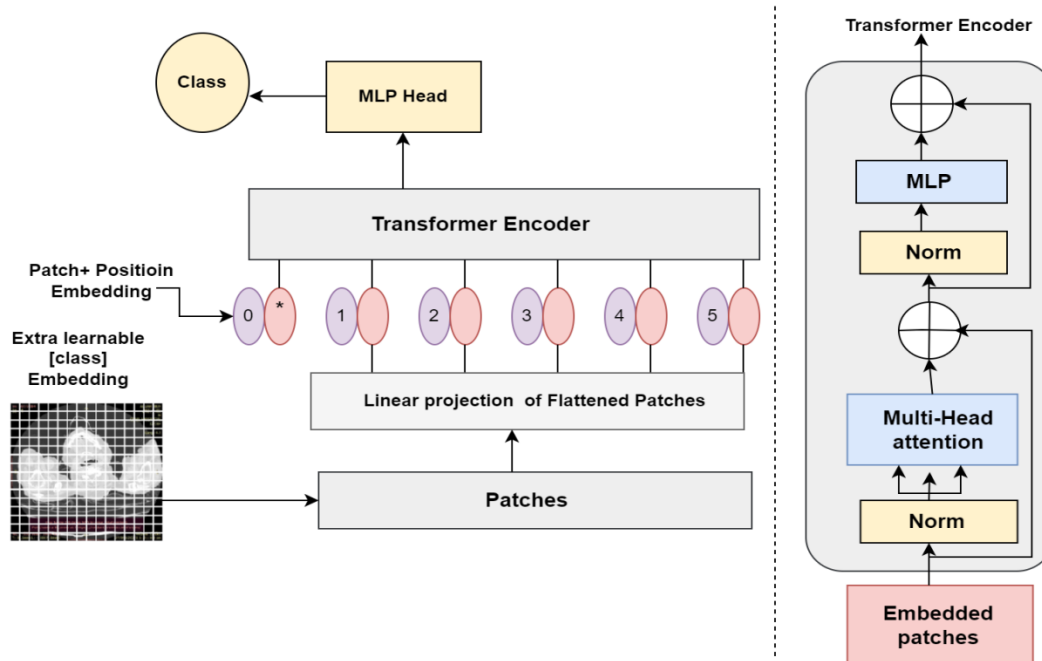


Figure 5-4 Proposed model vision transformer

Table 15 Parameters detail of vision transformer

Parameters	values
Learning rate	0.001
Weight decay	0.0001
Batch size	32
Num epochs	10
Image size	224
Patch size	14
Num patches	calculated
Projection dimension	64
Num heads	4

Transformer units	[128, 64]
Transformer layers	8
MLP head units	[2048, 1024]
Input shape	(224, 224,3)

5.9 Evaluation matrices

Each of the performance indicators (accuracy, precision, recall, confusion matrix and F1 score) is used to validate the expected output of brain tumor disease detection as shown in Equation 2.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

Where:

The fraction of instances correctly classified in the test set is called accuracy.

As a result, the tumor or not samples are represented by P and the negative or tumor vs not samples are represented by N as presented below in equations (3) and (4). The specificity and sensitivity formula is given by:

$$Specificity = \frac{TN}{TP + FN} \quad (3)$$

$$Sensitivity = \frac{TP}{TP + FP} \quad (4)$$

Precision: Number of instances correctly classified as positive/Total number of instances predicted as positive by the classifier, Equation 5. The precision is the ratio of the number

of samples correctly classified to the number of samples the classifier is expected to classify successfully.

$$Precision = \frac{TP}{TP + FP} \quad (5)$$

Recall is the proportion of true positive examples in the test set to all examples that are positive in the test set (Equation 6).

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

F1 score is the harmonic mean of precision and recall as shown in equation (7) which gives a combined measure of precision and recall.

$$F1\ score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (7)$$

The performance of the brain tumor detection model is tested using evaluation metrics. “Accuracy” displays the number of cases correctly predicted by the model. Precision accounts for how many of the cases that were predicted were correct. The recall is defined as the number of real tumor cases that could be successfully identified by the model. Specificity is how accurate the system is in identifying patients without brain tumors. A confusion matrix makes it easy to compare the correct and incorrect predictions in a table. F1 score is a combination of precision and recall and provides an overall performance result.

Chapter 6

Results & Discussions

6.1 Introduction

The results of the assessment of the suggested method for identification brain tumor diseases are shown in this section. In this section, the efficiency of the suggested brain cancer diseases identification design has been approved using several types of evaluation criteria and comparisons with other models.

6.2 Training and Testing Set

The dataset for brain tumor diseases identification comprises a total of 6,674 images. In the context of model development, 4,815 images are allocated for training, providing a diverse and representative set for the deep learning algorithm to learn patterns associated with both fibrosis and consolidation. This training set serves as the model's basis to acquire the necessary knowledge and optimize its parameters. Subsequently, a separate testing set of 1,859 images is set aside for evaluation and validation of the trained model's performance. This independent set ensures an impartial assessment of the algorithm's generalization capabilities, simulating real-world scenarios where the algorithm encounters unseen data, algorithm's accuracy, sensitivity, specificity, and overall efficacy in accurately identifying brain tumor diseases.

6.3 Performance Measurement

In this study, the performance of CNN architecture, Inception V3, ResNet50, MobileNet, Swin transformer and a proposed method Vision_transformer were examine, each assessed for its accuracy in distinguishing between different diseases affecting lungs. The contrast shows that the recommended course of action is more accurate.

6.3.1 CNN Architecture

The outcomes of the CNN architecture in brain tumor disease detection show off its effectiveness in advancing the cutting edge of medical imaging research. The CNN architecture's success in accurately identifying brain tumor disease underscores its potential as a valuable tool for early diagnosis and intervention, helping to achieve better results for patients in the realm of respiratory health.

In the instruction stage, the model refines its variables to accurately classify instances within the dataset containing brain tumor disease identifying images. The training accuracy metric gauges the model's ability in detecting these instances, as shown in figure. The validation phase assesses the evaluation process and the ability of the

framework to be applied to new, untested inputs accuracy indicates its execution on independent samples of brain tumors of both high and low grades grade cases.

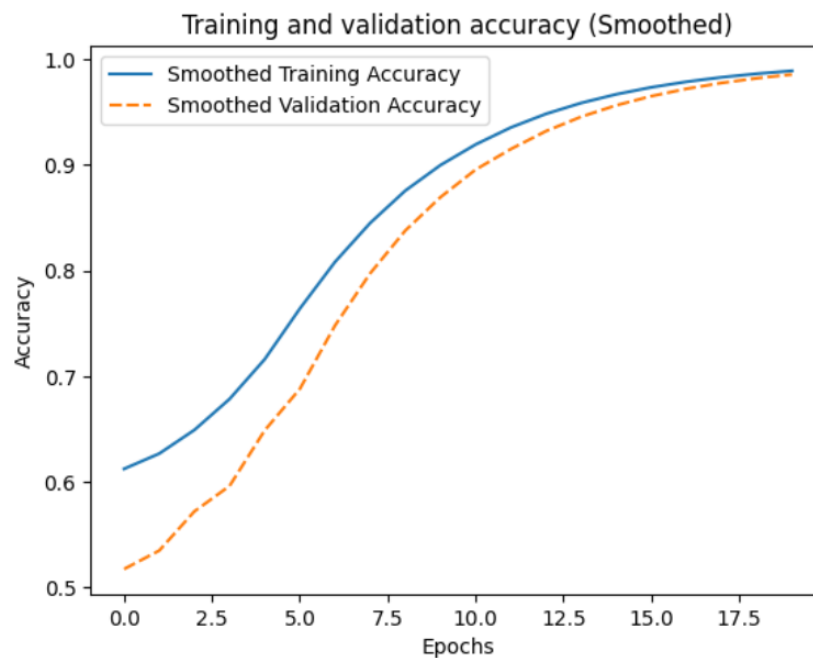


Figure 6-1 Training and validation accuracy of CNN architecture

The variance between expected and actual information is measured by the validation loss during validation, as shown in figure. High precision of training and low training and validation losses indicate effective learning and generalization, showcasing the CNN architecture's capability to discern and classify patterns associated with brain tumor accurately.

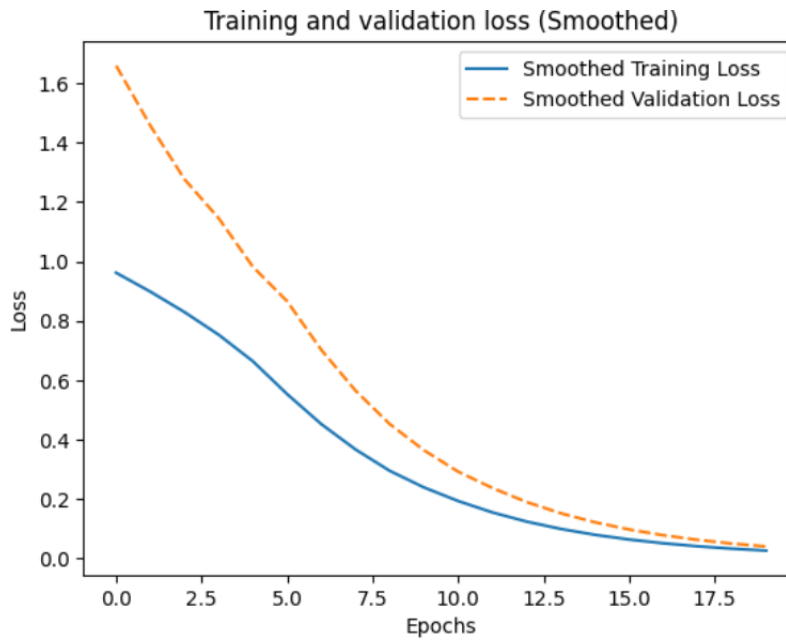


Figure 6-2 Training and validation loss of CNN architecture

The confusion matrix is used for the assessment of the classifying accuracy of the model, as shown in Figure 6.3. The confusion matrix for the CNN architecture used in identifying brain tumor disease, with a specific emphasis on high-grade and low-grade tumors, gives a thorough evaluation of the algorithms ability to classify these two categories.

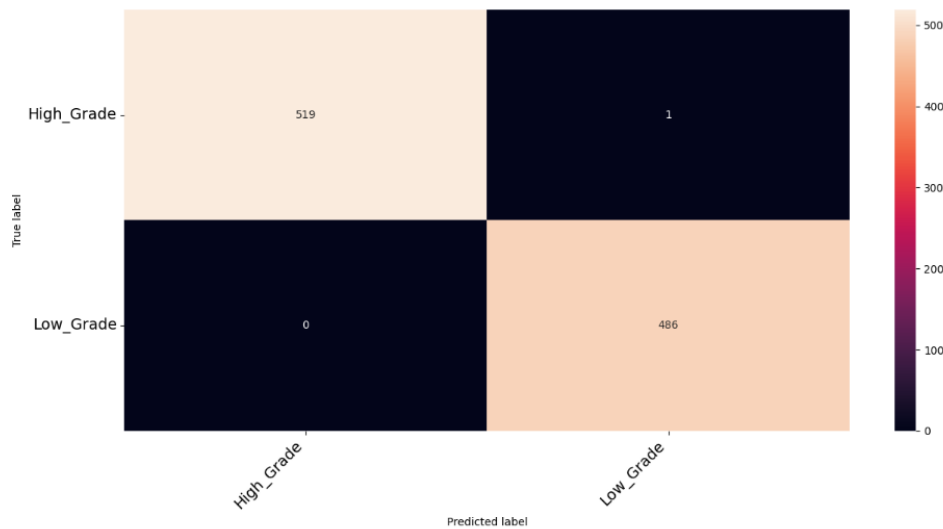


Figure 6-3 Confusion matrix of CNN architecture

The performance of the CNN architecture in the detection of brain tumor diseases, particularly distinguishing between high-grade and low-grade tumors, is effectively evaluated using the metrics of precision, recall, and F1 score. Precision evaluates the model's success in accurately identifying true positive instances while reducing false positives. Recall, or sensitivity, measures the model's capacity to correctly find every essential case, aiming to lower false negatives. The F1 score harmonizes precision and recall into a single measure that accounts for incorrect results as well as false positives. For the detection of high and low-grade tumors, these metrics together provide a holistic view of the CNN architecture's proficiency in precisely categorizing cases of these distinct brain tumor conditions. High precision indicates a low rate of misclassifying non-diseased instances as positive, high recall reflects effective identification of actual disease cases and a high F1 score signifies a balanced performance between recall as well as accuracy. These metrics guide the refinement of the CNN architecture for improved diagnostic accuracy in high and low grade brain tumor identification.

Table shows the results, which summarize the accuracy, accuracy and recollection generated from the confusion matrix obtained by the proposed CNN model.

Table 16 Performance matrices of CNN architecture

	F1- score	Precision	Recall	Support
0	0.97	0.99	0.98	520
1	0.96	0.98	0.97	486
Accuracy	0.99			1006
Macro- avg	0.99	0.99	0.99	1006
Weighted -avg	0.99	0.99	0.99	1006

6.3.2 ResNet-50

Regardless, the inherent strengths of ResNet-50, particularly its deep residual learning architecture designed to capture intricate features in medical images, there may be instances where the model struggles to achieve the desired accuracy in discerning respiratory conditions. ResNet-50 is designed to capture intricate features associated with respiratory conditions. In the learning stage, the model learns from a dataset comprising images of brain tumor identification, optimizing its parameters to enhance accuracy. The training accuracy metric reflects the appearance of the model proficiency in correctly classifying instances examples within training dataset, as shown in figure. The validation phase assesses ResNet-50's the capacity to extrapolate to fresh, untested data, with the validation accuracy indicating its performance on independent samples.

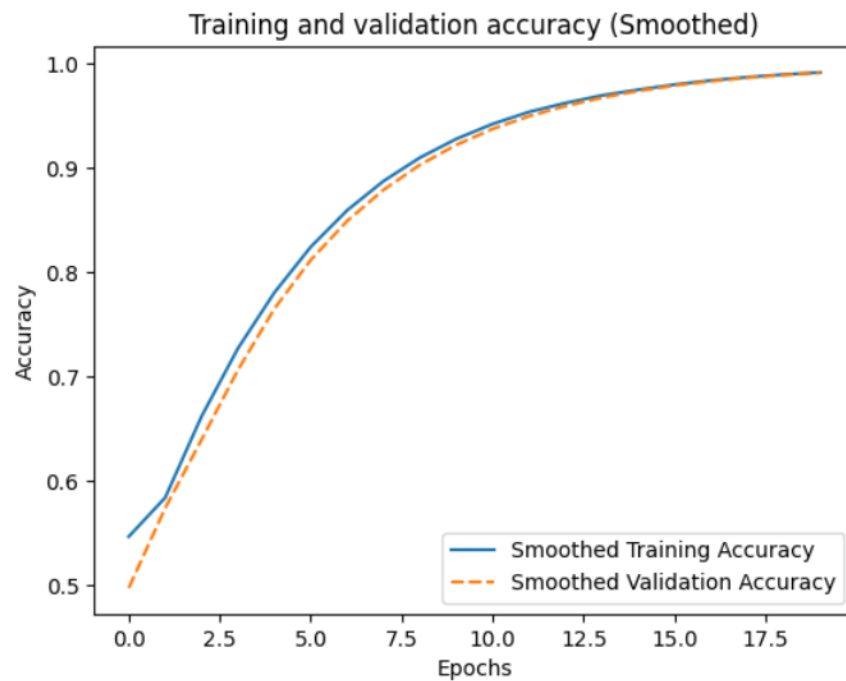


Figure 6-4 Training and validation accuracy of ResNet-50

The training loss quantifies the optimization process. The Measured as the difference between expected and actual values, loss of validation during validation. The training and validation loss of ResNet-50 is shown in figure.

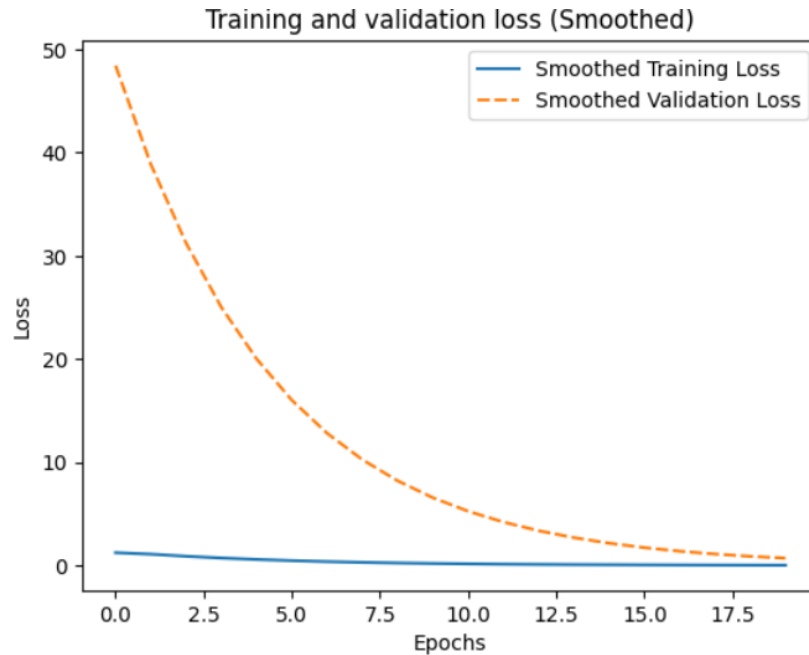


Figure 6-5 Training and validation loss of ResNet-50

Table shows that the ResNet50 model effectively distinguishes between two classes (0 and 1) in a dataset. Its precision for class 0 is perfect (0.99), but recall is 0.88. For class 1, recall is perfect (0.99), but precision is slightly lower (0.91). The model's overall accuracy is 0.94, with macro and weighted averages providing insights into performance across both classes.

Table 17 Performance matrices of ResNet-50

	F1-score	Precision	Recall	Support
0	0.93	0.97	0.88	520
1	0.95	0.91	0.95	486
Accuracy	0.94			1006
Macro-avg	0.94	0.95	0.94	1006
Weighted -avg	0.94	0.95	0.94	1006

The study evaluates ResNet-50's ability to identification of brain tumor disease, specifically high and low grade, using a confusion matrix. True Positives (TP) indicate accurate predictions of brain tumor disease, True Negatives (TN) indicate non-disease cases, and False Positives (FP) occurs when the model incorrectly predicts disease. This analysis provides an insight into ResNet-50's performance in lung disease diagnosis is shown in figure.

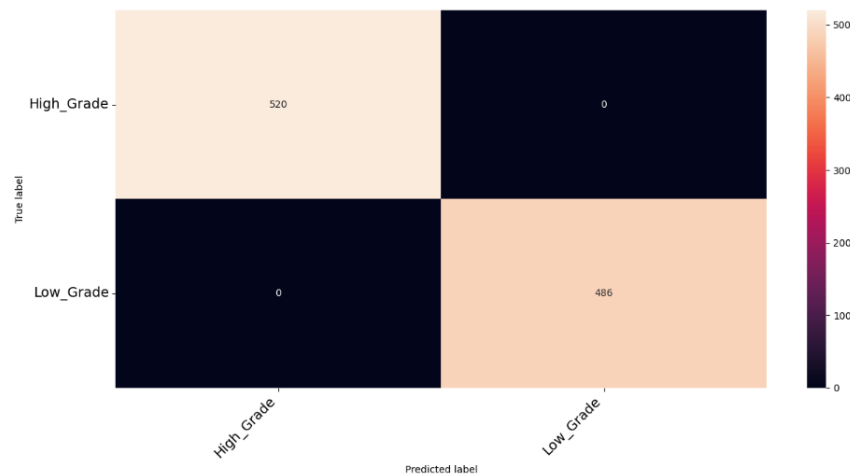


Figure 6-6 Confusion matrix of ResNet-50

6.3.3 Inception V3

In the evaluation of cranial tumor identification, the results obtained from Inception-V3 may not demonstrate superior performance when compared to other CNN architectures. Despite its unique architecture and multi-scale convolutional operations, Inception-V3 may face challenges in certain scenarios, exhibiting limitations in accurately capturing nuanced features associated with respiratory conditions. During the training phase, Inception-V3 learns from a dataset comprising images of lung diseases, optimizing its parameters to improve accuracy. The training accuracy metric signifies the model's proficiency in correctly classifying instances within the training dataset. The validation phase assesses the capacity of the framework to generalize to new, unseen data, as indicated by the validation accuracy, as shown in figure.

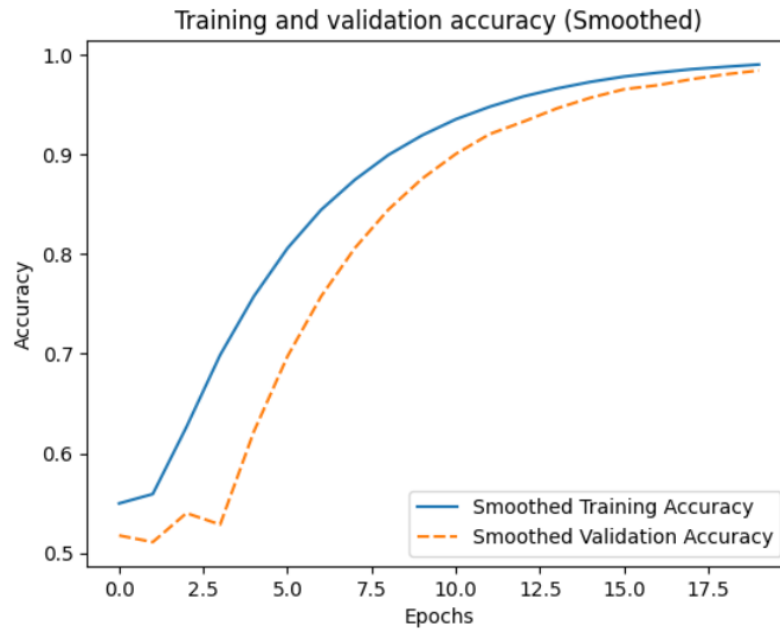


Figure 6-7 Training and validation accuracy of Inspection-V3

Figure show that the difference between expected and actual values during validation is measured as verification loss. High training accuracy and low training and validation losses are indicative of effective learning, suggesting Inception-V3's capacity to discern brain tumor disease patterns.

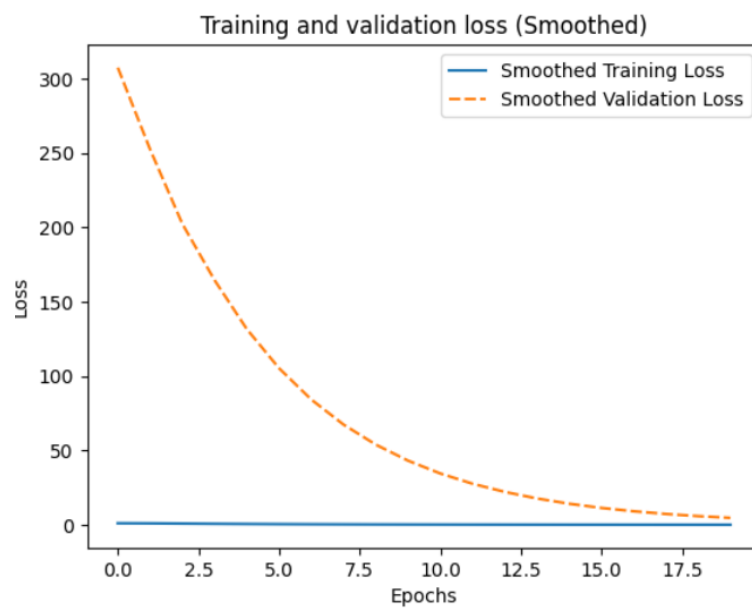


Figure 6-8 Training and validation loss of Inspection-V3

The study evaluates Inception-V3's ability to detect lung diseases, specifically fibrosis and consolidation, using a confusion matrix. True Positives (TP) indicate accurate predictions of brain tumor disease, True Negatives (TN) indicate non-disease cases, and False Positives (FP) occurs when the model incorrectly predicts disease. This analysis provides an insight into ResNet-50's performance in brain tumor identification appears in figure 4.9.

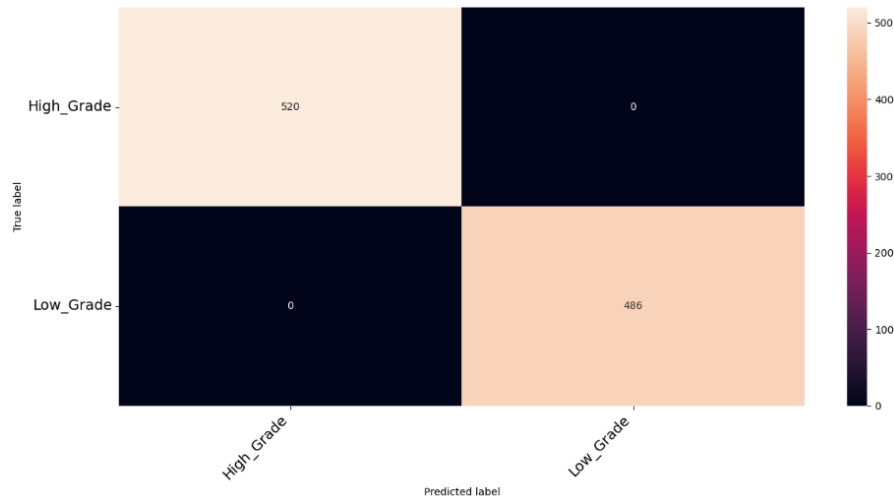


Figure 6-9 Confusion matrix of Inspection-V3

Table shows the Inception-V3's performance in lung disease detection is impressive, with high values for recall, accuracy, and the F1 score. The model achieves 99% precision for non-disease instances and 98% recall for lung disease instances. It also accurately classifies all actual disease cases, with a F1-score of 99%. Overall, the model achieves 99% accuracy across both classes.

Table 18 Performance matrices of Inspection-V3

	F1-score	Precision	Recall	Support
0	0.98	0.96	0.98	520
1	0.99	0.99	0.97	486
Accuracy	0.99			1006

Macro- avg	0.99	0.96	0.97	1006
Weighted -avg	0.99	0.98	0.98	1006

6.3.4 Mobile-Net

The visualization of precision as well as loss in instruction and verification graphs for MobileNet-based brain tumor identification offers a thorough insight into the model's performance dynamics. Figure shows the graphical representations provide a visual narrative of the model's learning trajectory throughout the instruction process. In the course of training, MobileNet is exposed to a diverse dataset containing brain tumor high order and low order images, allowing the model to learn intricate patterns and features associated with various brain tumor diseases. The precision to instruct reflects how well the model has adapted to the instruction set. High validation accuracy signifies the generalizability of the model well to new.

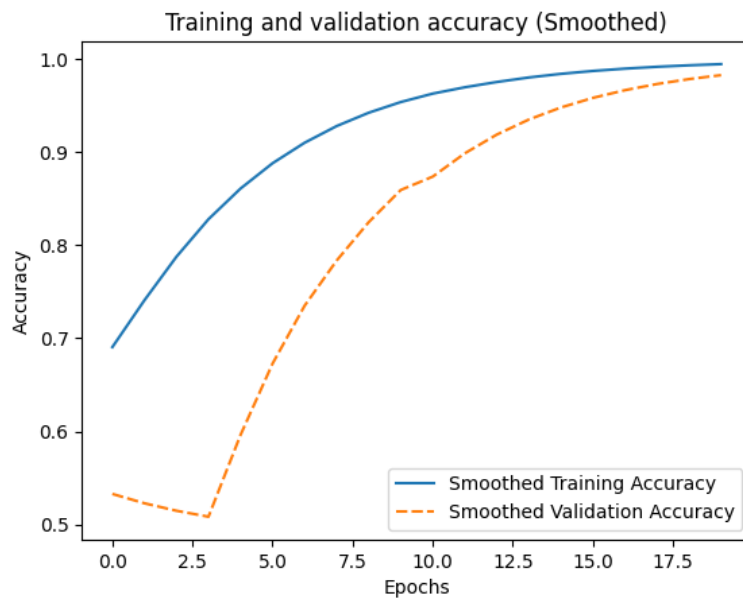


Figure 6-10 Training and Validation accuracy of Mobile-Net

The training and validation loss graph illustrates the framework's capacity for generalization during the training process, depicting the gradual decrease in error as it

refines its predictions on the brain tumor disease dataset. The validation loss graph is instrumental in evaluating the capacity of the model to generalize to fresh, untested data. A decreasing trend in validation loss implies that the learning set is not being overfitted by the model and is capable of making accurate predictions on diverse datasets, as shown in figure.

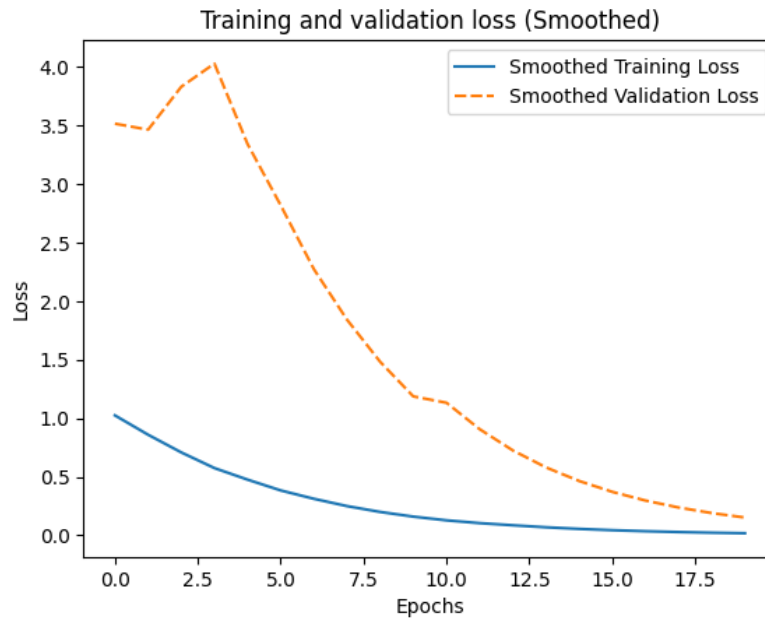


Figure 6-11 Training and validation loss of Mobile-Net

MobileNet's classification results for lung disease detection show high performance, with precision of 97% for non-disease instances and 99% recall for brain tumor disease instances. The model's F1-score is 98% for class 0, and 99% for class 1, indicating accurate classification of brain tumor disease identification and non-disease instances, as shown in table 4.5. These high precision, recall, and F1-score values highlight MobileNet's effectiveness in accurately identifying brain tumor disease and non-disease instances.

Table 19 Performance Matrics of Mobile-Net

	F1-score	Precision	Recall	Support
0	0.98	0.97	0.99	520

1	0.99	0.91	0.98	486
Accuracy	0.99			1006
Macro- avg	0.99	0.98	0.94	1006
Weighted -avg	0.99	0.99	0.99	1006

6.3.5 Swin-Transformer

The training and verification accuracy graphs for the Swin Transformer model are pivotal components of the evaluation process. These visual representations offer a nuanced understanding of the model's learning dynamics in the instruction stage, as shown in figure. The Swin Transformer's training and validation accuracy graphs reveal its learning dynamics. A rising curve in the training graph shows that the model is capable of learn brain tumor disease patterns, while a sustained upward trend in the validation graph indicates its robust performance on diverse datasets.

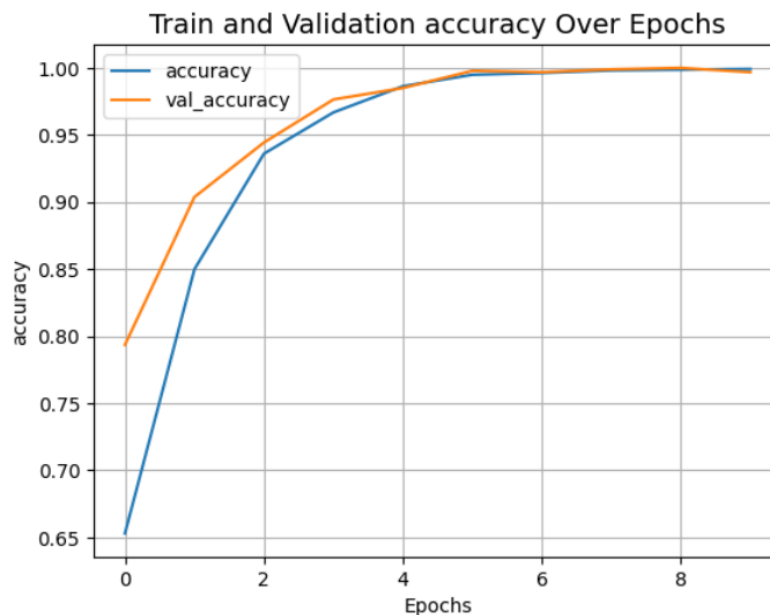


Figure 6-12 Training and validation accuracy of Swin transformer

The lack of verification and training graphs serve as critical tools for assessing the model's training dynamics and generalization capabilities, as shown in figure. The training loss graph illustrates the diminishing trend of predictive errors as the model refines its understanding of complex patterns within the brain tumor dataset over successive epochs.

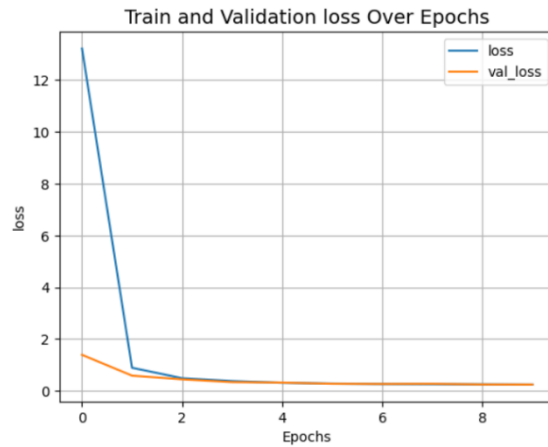


Figure 6-13 Training and validation loss of Swin transformer

The Swin Transformer's brain tumor disease identification model shows exceptional accuracy and reliability in table. It achieves 99% precision for non-disease instances, 98% recall for brain tumor lung disease instances, and 99% overall accuracy for brain tumor instances. The model's high precision, recall, and F1-score values demonstrate its robustness in distinguishing between lung disease and non-disease instances.

Table 20 Performance matrices of swin transformer

	F1-score	Precision	Recall	Support
0	0.97	0.99	0.98	520
1	0.98	0.96	0.97	486
Accuracy	0.99			1006
Macro-avg	0.99	0.98	0.96	1006
Weighted -	0.99	0.99	0.99	1006

avg				
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The efficacy of the Swin Transformer in identifying brain tumor diseases, particularly high and low grades, is assessed through a confusion matrix that features four principal components: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), as depicted in figure. Analyzing these components aids in quantifying the model's capacity to differentiate between instances of brain tumor diseases and those without the disease.

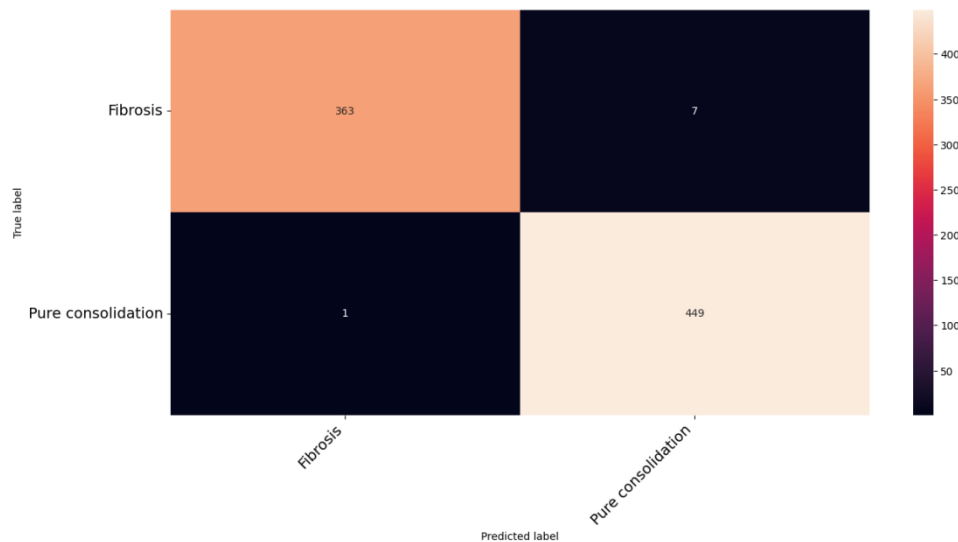


Figure 6-14 Confusion matrix of CNN architecture

6.3.6 Proposed Vision_transformer with Self-attention

The Proposed ViT with self-attention's performance in brain tumor identification is evaluated through training and testing accuracy. Training accuracy measures the model's adaptation to the practice information, while accuracy of assessment assesses its generalization capabilities. High training accuracy indicates the model's proficiency in capturing nuances, while comparable validation accuracy suggests it can generalize well to diverse instances, as shown in figure. Comprehensive analysis of these accuracy values is crucial for ensuring accurate brain tumor identification.

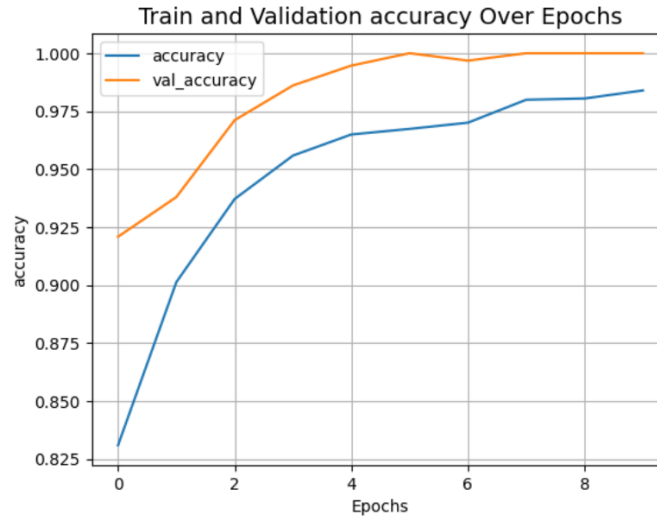


Figure 6-15 Training and validation accuracy of proposed model

The evaluation of the Proposed Vision Transformer with self-attention within the field of brain tumors diseases identification encompasses a thorough examination of training and validation loss. These key metrics serve as indicators of the model's convergence and generalization during the learning process, as shown in figure. The training loss graph illustrates the model's progression toward optimal performance over successive epochs.

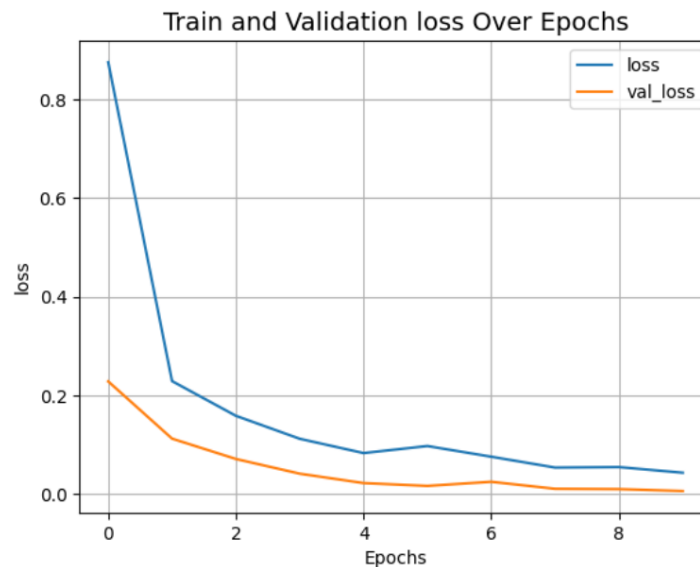


Figure 6-16 Training and validation loss of proposed model

The results of classification of Proposed Vision Transformer with self-attention in brain tumor disease identification are presented in table. In the case of class 0 (which contains

images of non-disease in it), the level of accuracy can be as high as 99%, meaning that 99% of the predictions for the non-disease class are correct. At 99% the recall for class 0 indicates that the model correctly identifies all true negative cases. The F1-score for class 0 is 99%, which is a balanced reminder and accuracy rate. For class 1 (which is the cases of lung disease), the precision is 99%, meaning that all the predicted cases of lung disease are correct. The model has a recall of 99%, which means that it can detect the majority of actual disease cases. The F1 score of class 1 is 99%. The model has a remarkable accuracy of 99.61%, supporting its power to accurately classify occurrences in the dataset. The accuracy, recall, and F1-score all indicate the model's robustness and correctness in differentiating between brain tumor disease and non-disease cases, proving the effectiveness of the proposed Vision Transformer (VT) with self-attention.

Table 21 Performance matrices of proposed model

	F1-score	Precision	Recall	Support
0	0.99	0.99	0.98	520
1	0.99	0.99	0.99	486
Accuracy	0.99			1006
Macro-avg	0.99	0.99	0.99	1006
Weighted - avg	0.99	0.99	0.99	1006

The Proposed Vision Transformer's ability to identify brain tumor disease, including high grade and low grade, is evaluated using a confusion matrix, as shown in figure 4.17. This matrix consists of four components: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). Analyzing these values provides insights into the Transformer's strengths and areas for improvement in detecting lung diseases.

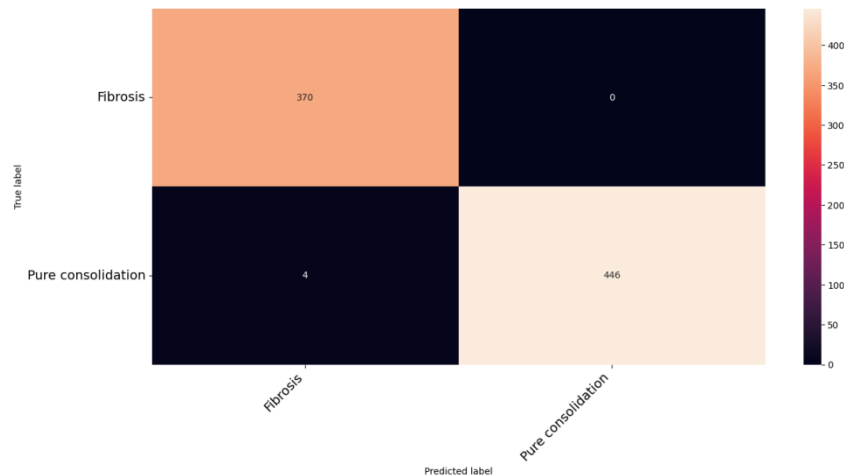


Figure 6-17 Confusion matrix of proposed model

Limitations

Interpretability is limited, making it difficult to decipher specific features. Data dependency and potential class imbalances in datasets also pose challenges. A Vision Transformer with self-attention achieved 99.47% lung disease detection accuracy, but potential dataset bias is a concern. Using comprehensive and representative datasets ensures model reliability across diverse scenarios, especially when dealing with diverse patient populations or imaging conditions. Overfitting, a risk where abnormal model accuracy is due to recall of training data has led to a lack of robustness and hindered clinical performance. Interpretation of the vision transformer, particularly with attention mechanisms, is a challenge due to its complex neural network architecture, requiring increased confidence and clinical adoption.

Future work

Future research could focus on improving the interpretability of the model by developing techniques to elucidate the specific features or patterns driving its decisions. To reduce data dependency and potential class imbalance, researchers will explore innovative data augmentation techniques and strategies to create more balanced data sets. Ensuring diversity in the training data and implementing techniques to remove biases can contribute to more robust and fair model performance. Optimizing CNN architectures for resource efficiency and collaborating between computer scientists and healthcare professionals is crucial for technologically sound and practical advancements in lung disease detection.

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